

The Gender Gap in Confidence: Expected but Not Accounted For[†]

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We investigate how the gender gap in confidence affects the views that evaluators (e.g., employers) hold about men and women. We find the confidence gap is contagious, causing evaluators to form overly pessimistic beliefs about women. This result arises even though the confidence gap is expected and even though the confidence gap shouldn't be contagious if evaluators are Bayesian. Only an intervention that facilitates Bayesian updating proves (somewhat) effective. Additional results highlight how similar findings follow even when there is no room for discriminatory motives or differences in priors because evaluators are asked about arbitrary, rather than gender-specific, groups. (JEL D82, D83, D91, J16, J22, M51)

Women are underrepresented and underpaid in many areas of the labor market, especially in male-stereotyped fields (Bertrand, Goldin, and Katz 2010; Goldin 2014; Blau and Kahn 2017; Michelmore and Sassler 2016). A large body of work has identified factors that may contribute to these gender gaps. Review articles highlight gender differences in the willingness to negotiate (Hernandez-Arenaz and Iriberry 2019) and compete (Niederle and Vesterlund 2011; Niederle 2016), gender differences in risk preferences (Croson and Gneezy 2009), and the role of discrimination (Riach and Rich 2002). Recent papers further narrow in on factors such as female leaders being rewarded less than equally effective male leaders (Grossman et al. 2019), women requesting lower starting salaries than men (Roussille 2021), women being less likely to self-report qualifications (Murciano-Goroff 2021), and women negotiating less even in a female-dominated profession (Biasi and Sarsons Forthcoming).

One of the literature's most robust findings is the gender gap in confidence (Lundeberg, Fox, and Punócohař 1994; Mobius et al. 2022), even among elite academics (Sarsons and Guo 2021) and especially in male-stereotyped fields (Beyer 1990; Bordalo et al. 2019; Coffman, Collis, and Kulkarni 2019b; Exley and Kessler 2022). Many papers highlight how the confidence gap may affect the “supply” of

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women in the labor market. For example, the confidence gap relates to women having lower earnings expectations (Reuben, Wiswall, and Zafar 2017), being less likely to enter competitive fields (Niederle and Vesterlund 2007; Buser, Niederle, and Oosterbeek 2014), being less likely to speak up (Coffman 2014), and being less likely to apply for challenging work (Coffman, Collis, and Kulkarni 2019a). But, less is known about how the confidence gap affects the “demand” for women, which is the focus of this paper.

How the confidence gap may affect the demand of women is unclear. On one hand, if others expect the confidence gap—perhaps due to movements such as “Lean In”—then they may account for it in a way that ensures that women’s relative under-confidence does *not* cause overly pessimistic beliefs about women. On the other hand, if others—such as employers, colleagues, and peers—do not expect or do not account for the confidence gap when forming beliefs about men and women, then the confidence gap will be “contagious.” For instance, the confidence gap may cause others to form overly pessimistic beliefs about women when reviewing job applications in which the candidate discusses their own performance and ability, when making promotion decisions that are in part based off of self-evaluations, and when selecting leaders and team members based off their self-reported qualifications. More pessimistic beliefs about women may, in turn, contribute to worse outcomes for women and may exacerbate gender discrimination (Bohren, Imas, and Rosenberg 2019; Coffman, Exley, and Niederle 2021).¹ We conduct an experiment that investigates whether individuals expect and account for the confidence gap to test between these hypotheses.

To first establish that there is a confidence gap in our setting, “workers” complete a math and science test and then answer 17 self-evaluation questions about their performance on the test. Workers are incentivized to accurately answer each self-evaluation question. The confidence gap proves robust: across all 17 self-evaluation questions—and significantly so in 16 of these questions—female workers provide more pessimistic beliefs about their performance than equally performing male workers do. For instance, when focusing on our main sample of workers for which there is no actual gender difference in performance, answers to our main self-evaluation question reveal that 80 percent of women believe they have a “poor performance” (i.e., a performance that is indicative of poor math and science skills) while only 56 percent of men do. Workers know that they are classified as having a poor performance if another randomly selected participant who does not know their gender deems the number of questions they got right on the test as indicative of poor math and science skills.

Then, to investigate how this confidence gap affects others’ beliefs about men and women, we incentivize “evaluators” to provide accurate beliefs about workers’ performance both before and after they learn how workers answer the main self-evaluation question. Specifically, after evaluators learn whether they will be asked to provide beliefs about a randomly selected male worker or instead a randomly selected female worker (who we refer to as “their worker”), the *Baseline*

¹ There are also many other important factors, e.g., the relative weight placed on luck (Erkal, Gangadharan, and Koh 2021).

treatment involves five main stages.² First, we elicit evaluators' *prior* by asking them to guess the percent chance that their worker has a poor performance. Second, we provide evaluators with *accurate* aggregate information about workers' self-evaluations: evaluators who are asked to provide beliefs about a randomly selected female worker are informed that 80 percent of female workers thought they had a poor performance, and evaluators who are asked to provide beliefs about a randomly selected male worker are informed that 56 percent of male workers thought they had a poor performance. Third, to examine how this information influences evaluators' beliefs, we elicit their *posterior* about the percent chance that their worker has a poor performance. Fourth, to investigate whether the confidence gap is expected, we elicit evaluators' beliefs about their worker's *overconfidence* and *underconfidence* by asking them to guess the percent chance that their worker is overconfident conditional on having a poor performance and the percent chance that their worker is underconfident conditional on having a "good performance." Finally, evaluators answer additional incentivized questions that measure their susceptibility to cognitive biases.

According to their prior beliefs, i.e., before receiving any information on workers' self-evaluations, evaluators expect that female workers are slightly more likely than male workers to have a poor performance. However, this expected performance gap is small (~ 3.9 percentage points) and is not statistically different from the true gap (~ 1.7 percentage points).

After evaluators receive information on workers' self-evaluations—information that conveys more pessimistic views held by female workers or more optimistic views held by male workers—does this expected performance gap become substantial because evaluators fail to account for the confidence gap in these self-evaluations? That is, does the confidence gap prove to be "contagious"? Or, is the potentially detrimental impact of the confidence gap avoided because evaluators expect and account for the confidence gap?

Two results *seem* to point towards the latter at first blush. First, as indicated via their beliefs about workers' confidence, evaluators expect the confidence gap in self-evaluations. Evaluators expect that, among workers with a poor performance, male workers are 8.25 percentage points significantly more likely than female workers to be overconfident and incorrectly guess that they have a good performance. Evaluators also expect that, among workers with a good performance, female workers are 10.07 percentage points significantly more likely than male workers to be underconfident and incorrectly guess that they have a poor performance. Second, we can calculate—from evaluators' priors, the information on workers' self-evaluations, and evaluators' beliefs about the accuracy of that information (given their beliefs about workers' confidence)—the posterior beliefs that evaluators would hold if they were Bayesian. These *implied Bayesian posterior beliefs* indicate that the confidence gap should not be contagious, and specifically, that the information on workers' self-evaluations should not result in overly pessimistic views about women.

We nonetheless find the opposite to be true: the confidence gap in workers' self-evaluations is contagious. After receiving information on workers'

² As explained in Section IC, we ask about a subgroup of workers for whom there are no actual gender differences in performance. But, as shown in Sections VB, VC, VI, and VJ, our results are not reliant on this restriction.

self-evaluations, evaluators hold an overly pessimistic view about the relative performance of women. According to their posteriors, evaluators now expect a large and statistically significant performance gap (~ 10.5 percentage points). This expected performance gap is indeed 6 times larger than the true performance gap and nearly 3 times larger than the gap in evaluators' priors. Thus, the confidence gap exacerbates the expected performance gap, even though Bayesian updating implies that it shouldn't and even though the confidence gap is expected.

In considering what prevents evaluators from accounting for the confidence gap when forming their posterior beliefs, one possibility relates to an "attention" problem: evaluators may simply fail to attend to the confidence gap when forming their posterior beliefs. To investigate this possibility, we test a light-touch intervention. In the *Attention* treatment, we make beliefs about confidence more salient by eliciting evaluators' confidence beliefs before—rather than after—their posterior beliefs. This intervention proves ineffective: the expected performance gap remains at the same (substantial and significant) level.

Another possibility relates to a "calculation" problem: evaluators may be either unable or unwilling to do the necessary calculations and Bayesian updating required to accurately account for the confidence gap. To investigate this possibility, we test a much more extensive intervention. In the *Calculation* treatment, to alleviate any difficulty with Bayesian updating, we provide evaluators with their implied Bayesian posterior beliefs before eliciting their posteriors. This intervention proves effective: the expected performance gap shrinks and is only marginally significantly different from the true performance gap. As additional evidence of the calculation problem, we also see that the extent to which evaluators' posteriors disfavor women is positively and significantly correlated with evaluators exhibiting base rate neglect.

To further investigate the calculation problem—and the extent to which our results are specific to gender—we examine whether similar results follow when we ask evaluators about arbitrary, rather than gender-specific, groups. In particular, in the *Unknown Gender* treatments, we ask evaluators about either "group-1" or "group-2" workers. Evaluators know that workers are assigned to these groups based on their answers to a question in a follow-up survey, but they do not know what this question is (and in particular, do not know that the question is about the worker's gender). This maintains the confidence gap between these two groups of workers while allowing differences between evaluators' posterior beliefs and their implied Bayesian posterior beliefs to reflect failures in Bayesian updating but not any gender-specific biases. We find that the results from these treatments are indistinguishable from the results where evaluators know the workers' gender, which shows that the confidence gap is contagious even when it is a calculation problem about arbitrary groups and cannot reflect discriminatory motives or differences in priors.

To summarize, our main results show that the confidence gap is contagious—causing the expected performance gap in which evaluators have overly pessimistic beliefs about women relative to men—and specifically point towards the role of failures in Bayesian updating rather than other sources such as taste-based discrimination against women *per se*. Three results support this conclusion, the latter two of which are explained in detail above. First, counter to what one may expect but consistent with prior findings in Card et al. (2020) in which the gender of the referee does not significantly affect the relative assessment of economics papers written by

men versus women, we find that female evaluators are just as likely as male evaluators to hold posterior beliefs that significantly disfavor women. Second, the only intervention that is somewhat successful at shrinking the expected performance gap is the *Calculation* treatment that assists evaluators with Bayesian updating. Third, the expected performance gap persists in the *Gender Unknown* treatments, i.e., the treatment in which we remove gender labels so the expected performance gap may reflect failures in Bayesian updating but not discriminatory motives against women.

In addition to the robustness of the expected performance gap evident from the above, we conclude with a few more notes on robustness. First, since how we classify workers as having a “poor performance” may contribute to the complexity involved in evaluators forming their posterior beliefs, we show that our results are robust to using a simpler performance outcome in which evaluators provide beliefs about whether a worker has performance in the top half—see Section VA for more details. Second, and related to recent work on understanding experts’ beliefs (DellaVigna and Pope 2018a, b), we replicate our results among a pool of participants for whom one may posit this type of problem is less complex: professional evaluators who self-report hiring and managerial experience. Third, our results persist across a variety of types of evaluators and across a variety of conditions, including when evaluators know additional information on the workers and when workers face strategic incentives.

To better understand the potential impact of gender differences in the labor market, our work complements the aforementioned rich literature on how the confidence gap affects the decisions made by men and women *themselves* by additionally examining how the confidence gap affects *others’ beliefs* about men and women. Our work is thus related to the small but growing body of literature on how the confidence gap affects others’ decisions—and hence may relate to others’ beliefs—about men and women. This literature shows that the confidence gap conveyed via group interactions may relate to women being selected less frequently as leaders (Reuben et al. 2012), that the confidence gap conveyed via workers’ self-reported beliefs may explain why providing these self-reports to employers does not mitigate their male hiring preference (Reuben, Sapienza and Zingales 2014), and that the confidence gap conveyed with employees’ self-evaluations does not influence employers’ relative ratings of their male and female employees (Bohnet, Hauser, and Kristal 2022).

Relative to this literature, part of our main contribution lies in eliciting a variety of incentivized beliefs that allow us to cleanly document and narrow in on *why* individuals do not account for the confidence gap.³ Indeed, our evidence makes clear that it is not simply an attention problem and instead points towards a calculation problem. This connects our work to the extensive literature on errors in Bayesian updating (see Benjamin 2019 for a review). For instance, the fact that our evaluators react *too much* to self-evaluation information relative to the Bayesian posterior is consistent with a growing literature that shows overinference from weak signals (Edwards 1968; Augenblick, Lazarus, and Thaler 2023; Ba, Bohren, and Imas 2023). Our results also relate to early (Kahneman and Tversky 1972b, 1973; Grether

³Reuben, Sapienza, and Zingales (2014) have unincentivized belief data consistent with their findings. Reuben et al. (2012) and Bohnet, Hauser, and Kristal (2022) do not have belief data.

1980; Koehler 1996) and more recent (Esponda, Vespa, and Yuksel 2024) work on base-rate neglect, work that documents a relationship between non-Bayesian updating and cognitive uncertainty (Enke and Graeber 2023), and other related belief updating biases such as correlation neglect (Enke and Zimmermann 2019), learning from missing information (Enke 2020), and failure to unlearn from retracted signals (Gonçalves, Libgober, and Willis 2021). In addition, that errors in Bayesian updating can contribute to worse beliefs about women relative to men also connects our findings to the work on discrimination that is reflective of *inaccurate* beliefs (Bordalo et al. 2019; Bohren et al. 2019). More generally, our results point towards the need for more extensive interventions that directly help individuals with Bayesian updating in order to account for the confidence gap. We discuss possibilities along these lines and other directions for future work in Section VI.

I. Experimental Design

A. Design Overview

Our experimental design involves two main types of participants: “workers” and “evaluators.” The workers are incentivized to accurately answer self-evaluations about their performance on a test, and the evaluators are incentivized to accurately provide beliefs about the workers before and after the evaluators are given information on the workers’ self-evaluations. In this way, our design is akin to a situation where evaluators (such as managers) hold beliefs about their workers and these beliefs may then be affected by self-evaluations that workers provide (such as in interviews and performance reviews).

An important question in our experimental design is what type of self-evaluation to focus on. Should we examine self-evaluations in which workers are asked about their absolute performance (e.g., the number questions they got right on the test), their relative performance (e.g., whether their test performance was in the top half among all workers), or their subjective performance (e.g., whether their test performance was “poor”)? Absolute, relative, and subjective performance outcomes all have been used in prior work and are relevant in many contexts outside of the laboratory. Thus, our design approach is three-fold.

First, our *Worker Study* asks workers to complete 17 self-evaluations about their performance, including absolute, relative, and subjective performance questions. This allows us to examine whether there are gender differences in self-evaluations—i.e., whether the confidence gap arises—in a wide range of different types of self-evaluations.

Second, our *Evaluator Study* asks evaluators to provide beliefs—incentivized for accuracy—about these workers both before and after being provided with information about just *one* type of self-evaluation. Specifically, to avoid confusing the evaluators and to allow for clean Bayesian benchmarks, our *Evaluator Study* elicits beliefs about our *main* self-evaluation. In choosing which self-evaluation should be our “main” self-evaluation, we chose a self-evaluation question in which workers provide beliefs about a *binary and subjective* performance outcome (detailed more in Section IB). The binary nature of this performance outcome facilitates Bayesian benchmarks, and the subjective nature of this performance outcome allows us to

build upon prior related work (Exley and Kessler 2022) and connect to important contexts outside of the laboratory where individuals complete self-evaluations and discuss their performance in more subjective ways.⁴

Third, we show that our focus on *one* type of self-evaluation in the *Evaluator Study* is not driving our results by documenting the robustness of our results in other study versions. In particular, all of our results—for both workers and evaluators—are robust to considering different self-evaluation questions, including a simpler measure of whether a worker’s performance is in the top half among other workers (see Section VA).

Below, to explain our main design most concisely, we will refrain from discussing these other self-evaluation questions for now and will instead focus on our *main* self-evaluation question. In particular, Section IB describes our *Worker Study*, with specific attention paid to the main self-evaluation question even though workers are asked to answer 17 self-evaluation questions. Section IC describes *Evaluator Study*, which only relates to our main self-evaluation question. Following this, Section ID briefly details our recruitment and implementation details, including for other studies. In total, we recruited 7,694 participants—mostly on Prolific and as detailed later.

B. Design for the Worker Study

The *Worker Study* involves two main parts: part 1 and part 2. In addition to a \$3 completion fee for a 15-minute study, workers may earn up to \$1 in bonus payment, randomly selected from either part 1 or part 2.

In part 1, workers answer a 10-question math and science test.⁵ Workers have 20 seconds to answer each question, and workers are never provided with any information on their performance on this test. If part 1 is selected as the part-that-counts, then workers earn \$0.10 for each question they answer correctly.

After part 1 but before part 2, workers report an answer 0–10 in response to the following (unincentivized) “classifier question.”⁶

- **Classifier Question:** An individual’s performance on the math and science test was indicative of poor math and science skills if the number of questions the individual answered correctly was less than or equal to ____.

In part 2, workers answer 17 self-evaluations—displayed in random order—about their own performance. If part 2 is randomly selected as the part-that-counts, then workers receive the amount they earn in one randomly selected self-evaluation and are incentivized to answer accurately.⁷ Online Appendix Table A.4 details all 17 of the self-evaluation questions, and the *main self-evaluation question* is as follows:

⁴The subjective nature also allows us to control information on *objective* performance (see Section VJ).

⁵We selected ten questions from the Armed Services Vocational Aptitude Battery (ASVAB), which is used to assess aptitude in various technical fields. We tell participants that “performance on this test is often used as a measure of cognitive ability by academic researchers.”

⁶Workers answered two classifier questions, but we focus here on the one that we use in the *Evaluator Study*. The full text of both questions can be found in online Appendix Table A.4.

⁷See the table note of online Appendix Table A.4 for details on randomization and incentives.

- **Main Self-Evaluation Question:** Did your classifier describe your performance on the math and science test as indicative of poor math and science skills?

In response to the main self-evaluation question, workers can select “yes” or “no” and know that they earn \$1 in that self-evaluation if their guess is correct. To answer the main self-evaluation question, workers are told that they will be matched with another worker (called their “classifier”) who is equally likely to be a male worker or female worker.⁸ We tell workers that their score is classified as “poor performance” if it was less than or equal to the threshold score that their classifier indicated in the Classifier Question described above. For example, if a worker’s classifier says that an individual’s performance is indicative of poor math and science skills if they answered 5 or fewer questions right, then that worker is classified as having a “poor performance” if they scored 0–5 on the test. While we will use this shorthand of “poor performance” throughout the rest of our paper for conciseness, we instead write out the definition of poor performance (performance on the math and science test that was indicative of poor math and science skills”) in the text of the questions provided to workers, as shown in online Appendix Table A.4.⁹ We will also use the shorthand of “good performance” to refer to the opposite.

C. Design for the Evaluator Study

In the *Evaluator Study*, evaluators are randomly assigned into one of six treatments. We will detail the *Baseline* treatment below, and we will describe the the additional five treatments later as they become relevant (see also online Appendix Figures A.1–A.3 for an overview of each of these treatments).

The *Baseline* treatment of the *Evaluator Study* elicits four beliefs: prior beliefs, posterior beliefs, underconfidence beliefs, and overconfidence beliefs. See online Appendix Table A.5 for the exact wording of the belief questions. In addition to a \$2 completion fee for a 10-minute study, evaluators may earn \$1 in bonus payment because they are incentivized to accurately provide these beliefs, as detailed below. Each of these four beliefs relates to whether “their worker” has a poor performance, defined in the same manner as noted above in Section IB.¹⁰ Evaluators know that their worker will be randomly selected from the available pool of female workers (and thus referred to as “your female worker”) or instead will be randomly selected from the available pool of male workers (and thus referred to as “your male worker”). Therefore, each evaluator is only asked about female workers *or* male workers.

To examine evaluators’ beliefs before they learn any information on workers’ self-evaluations, we first elicit an evaluator’s *prior belief* about the percent chance

⁸In the study, we actually refer to “classifiers” as “evaluators.” But, to avoid confusion with our later study versions, we refer to them as classifiers in our paper.

⁹Specifically, the main self-evaluation question corresponds to Self-Evaluation 8B in online Appendix Table A.4. In addition to the definition of poor performance being written out, note that the “classifier” is referred to as their “evaluator” as previously explained in Footnote 8.

¹⁰In the question text provided to evaluators, the definition of poor performance is written, and the worker’s “classifier” is referred to as the worker’s “evaluator” (see online Appendix Table A.5).

that their worker has a poor performance by asking them the “Prior Belief” question noted below.

- **Prior Belief:** What do you think is the percent chance that your male/female worker in this prediction had a poor performance?

Next, to examine how evaluators’ beliefs are influenced by information on workers’ self-evaluations, we provide them with *accurate* information on the workers’ self-evaluations and then elicit their *posterior beliefs*. Specifically, from the available pool of workers from which the worker could be randomly selected, evaluators are accurately informed that 80 percent of female workers thought they had a poor performance if their worker is a randomly selected female worker, or instead are accurately informed that 56 percent of male workers thought they had a poor performance if their worker is a randomly selected male worker. We then elicit evaluators’ *posterior belief* about the percent chance that their worker has a poor performance by asking them the “Posterior Belief” question noted below.

- **Posterior Belief:** After completing the math and science test, 56%/80% of male/female workers predicted that they had a poor performance. What do you think is the percent chance that your male/female worker in this prediction had a poor performance?

Finally, to assess how likely evaluators think it is that their worker is overconfident or underconfident, we elicit evaluators’ *overconfidence belief* and *underconfidence belief* by asking them, via a strategy-method style elicitation, the following “Overconfidence Belief” question and “Underconfidence Belief” question.

- **Overconfidence Belief:** If your male/female worker in this prediction had a poor performance, what do you think is the percent chance that he/she is overconfident because he/she predicted that he/she did NOT have a poor performance?
- **Underconfidence Belief:** If your male/female worker in this prediction did not have a poor performance, what do you think is the percent chance that he/she is underconfident because he/she predicted that he/she had a poor performance?

We conclude the main experimental design with two additional notes: one on the available pool of workers and another on incentives. On the available pool of workers, recall that evaluators provide beliefs about their male *or* female worker who is randomly selected from the available pool of workers. Evaluators are informed that this available pool of workers is the group of workers who had performances in the “middle,” or in the twenty-fifth–seventy-fifth percentile, in the *Worker Study*. This restricted worker pool allows us to ensure that there are no gender differences in the actual performance of workers, but it does introduce some complexity in terms of how we describe the available pool of workers to evaluators.¹¹ We thus emphasize

¹¹ Specifically, we describe this to evaluators as follows: “Workers who had performances in middle neither performed the best nor performed the worst. According to the number of questions they got right on the math and

that, as an important robustness check, we show that similar results persist when we remove this restriction and instead ask evaluators to provide beliefs about the full pool of workers in the *Evaluator (Full Distribution) Study* (see Section VB). We also note that other study versions show that similar results persist when we do not have to rely on this restriction to ensure there are no gender differences in the actual performance—see the *Evaluator (Professional Evaluators) Study* in Section VC, the *Evaluator (Additional Demographics) Study* in Section VI, and the *Evaluator (Known Performance) Study* in Section VJ.

On incentives, evaluators know they are equally likely to receive how much they earn from (i) their prior belief, (ii) their posterior belief, or (iii) either their overconfidence or underconfidence belief, depending on which of these two beliefs is relevant given the strategy-method elicitation. Evaluators report each belief in the form of a percent chance of some outcome being true (0–100 percent) and may earn a \$1 bonus according to an incentive-compatible Becker-DeGroot-Marschak (BDM) procedure.¹² In addition, at the end of the study after they have provided all of the above beliefs, participants are surprised with the opportunity to earn \$1 if they correctly answer one question, selected at random, out of five additional questions. These five additional questions test for common cognitive biases that might correlate with belief updating behavior (see Section IVA for details).

D. Implementation and Recruitment Details

In all of our studies, participants receive ample instructions and are required to correctly answer understanding questions before proceeding to the main parts of our study. Rather than excluding participants, they are given as many times as needed to correctly answer the understanding questions. For full experimental instructions of all study versions that we run, see the supplemental online Appendix.

For our *Worker Study*, we recruited 403 participants on Prolific to complete our study as “workers.”¹³ After excluding 10 participants who neither identify as men nor women because we are under-powered to consider this group, this resulted in 393 workers. For an overview of this study (referred to as the *Worker Study—Baseline Treatment*) as well as additional study versions that involve workers, see online Appendix Table A.1.

For the *Evaluator* study, we recruited 2,400 participants on Prolific to complete studies as “evaluators” (see footnote 13 for eligibility criteria). These evaluators were randomized into one of six treatments of our *Evaluator Study*: the *Baseline* treatment ($n = 402$), the *Attention* treatment ($n = 403$), the *Calculation* treatment

science test, workers who had performances in the middle performed better than or equal to at least one-quarter of all workers, and they performed worse than or equal to at least one-quarter of all workers.”

¹²Specifically, they are told that to secure the largest chance of earning \$1 from each self-evaluation, they should report their most-accurate guess. They are then allowed to click on a button to reveal the precise payment rule. For the 19 percent of participants who choose to reveal this information, they are provided with full details of the BDM procedure. For more on the BDM procedure, see Mobius et al. (2022). Future work may also examine the robustness of these results to instead eliciting beliefs as frequencies.

¹³To be eligible for our study, participants needed to have completed at least 100 prior submissions on Prolific with an approval rating of 95 percent or greater and chose the United States as their residence. Also, since we recruited a gender balanced sample, participants must have selected either Male or Female for their sex on the Prolific platform—although we use their self-identified gender from our follow-up survey.

($n = 405$), the *Baseline, Unknown Gender* treatment ($n = 405$), the *Attention, Unknown Gender* treatment ($n = 392$), and the *Calculation, Unknown Gender* treatment ($n = 393$). For an overview of these six treatments in our *Evaluator Study*, see online Appendix Table A.2. For an overview of additional study versions that involve evaluators, see online Appendix Table A.3.

In total, in addition to our *Worker Study* and our *Evaluator Study*, we recruited an additional 1,091 workers and 3,800 evaluators to complete additional study versions, which we will discuss as they become relevant in this paper.¹⁴ All of our data are available in the repository Exley and Nielsen (2024).

II. Worker Results

To establish the confidence gap, we first examine data from the *Worker Study*.

Table 1 presents results on how male and female workers answer the *main self-evaluation question* by showing the likelihood that a worker believes that they have a poor performance (the dependent variable equals 1 if a worker believes that they have a poor performance and 0 otherwise) regressed on *Female*, which is an indicator for female workers.

The estimates in column 1 show a clear confidence gap among the full pool of workers: 57 percent of male workers believe they have a poor performance (see the coefficient estimate on the constant) while 73 percent of female workers believe they have a poor performance (note the sum of the coefficient estimates on the constant and *Female*). This confidence gap arises despite the fact that the actual likelihood of a poor performance is 53 percent among female workers and 47 percent among male workers ($p = 0.09$).¹⁵ In addition, the inclusion of performance fixed effects in column 2 reveals that this confidence gap is statistically significant when comparing equally performing men and women.

The estimates in column 3 also show a clear confidence gap among the available pool of workers that evaluators are asked about (i.e., workers who had performances in the middle): 56 percent of male workers in the available pool of male workers believe they have a poor performance while 80 percent of female workers in the available pool of female workers believe they have a poor performance.¹⁶ This confidence gap arises despite the fact that the actual likelihood of a poor performance is 50 percent among these female workers and 48 percent among these male workers ($p = 0.56$). The inclusion of performance fixed effects in column 4 reveals that this confidence gap remains statistically significant when comparing equally performing men and women.

¹⁴Related to one of our additional worker and evaluator studies, we also recruited 100 participants to complete the study as “employers,” as detailed in footnote 35 in the online Appendix.

¹⁵To calculate a worker’s true chance of a poor performance, we determine the percent of classifiers who classified the worker’s score as indicative of poor math and science skills in response to the Classifier Question. Then, to determine the chance that a randomly selected male/female worker has a poor performance, we average these chances across all male/female workers.

¹⁶One might wonder whether this result arises from differences in beliefs about absolute performance or about differences in beliefs in the poor performance “standards” of the classifiers. Controlling for performance, or performance and beliefs about performance, we find no significant gender difference in how workers answer the classifier questions ($p < 0.1$). We do not elicit workers’ beliefs about the classifier thresholds of others.

TABLE 1—SELF-EVALUATIONS IN THE *BASILINE* TREATMENT OF THE *WORKER STUDY*

	Workers' answer to main self-evaluation question			
	All workers		Available pool of workers	
	(1)	(2)	(3)	(4)
<i>Female</i>	0.185 (0.047)	0.155 (0.044)	0.233 (0.057)	0.232 (0.056)
<i>Constant</i>	0.573 (0.035)		0.563 (0.044)	
Observations	393	393	249	249
Perf fixed effects	No	Yes	No	Yes

Notes: Standard errors are robust and shown in parentheses. Results are from OLS regressions of the responses provided to the main self-evaluation question, coded as 1 if the workers guess they have a “poor performance” and 0 otherwise. *Female* is an indicator for the worker identifying as a woman. Perf fixed effects are dummies for each possible performance out of the 10 questions on the test. In columns 1–2, data are from the 393 participants who identified as a man or a woman in the *Baseline* treatment of the *Worker Study*. In columns 3–4, data are further restricted to the available pool of workers that evaluators are asked about—i.e., male and female workers with performances in the “middle” or twenty-fifth–seventy-fifth percentile.

While the above focuses on documenting the confidence gap in response to the main self-evaluation question, recall that workers answered 16 other self-evaluation questions as well. Online Appendix Table B.1 presents the regression results of all self-evaluations. These results reveal that the confidence gap is robust in response to all 17 self-evaluation questions, and significantly so in 16 of the 17.¹⁷ Specifically, the column headers in online Appendix Table B.1 refer to the relevant self-evaluation question label that is detailed in online Appendix Table A.4. For instance, Column “0” of online Appendix Table B.1 presents the workers’ responses to Self-Evaluation Question “0” as labeled in online Appendix Table A.4.

Focusing on absolute performance outcomes, panel A of online Appendix Table B.1 shows that women believe they got fewer questions correct than men in absolute terms (column 0), believe that they are less likely than men to have answered at least 3 questions correctly (columns 1B and 1C), to have answered at least 5 questions correctly (columns 2B and 2C), and to have answered at least 7 questions correctly (columns 3B and 3C).

Focusing on relative performance outcomes, panel B of online Appendix Table B.1 shows that women are less likely than men to believe they scored in the top half relative to all other participants who took the study (columns 4B and 4C), relative to women who took the study (columns 5B and 5C), and relative to men who took the study (columns 6B and 6C).

Finally, focusing on the subjective performance outcomes, panel C of online Appendix Table B.1 shows that the results are robust to different types of subjective performance outcomes (columns 7B–8C).

¹⁷The confidence gap is not statistically significant in column 3C, statistically significant at the $p < 0.1$ level in column 3B, and statistically significant at the $p < 0.01$ in all 15 other columns. The results in columns 3B and 3C may in part reflect that even male workers thought it was very unlikely to have answered at least 7 questions correctly. Indeed, in response to the binary self-evaluation question in column 3B, only 13 percent of male workers thought they got 7+ questions right. In response to the percent chance self-evaluation question in column 3C, the average believed percent chance of getting 7+ questions right was 26 percent among male workers.

Thus, taken together, our results reveal that the gender gap in confidence persists when we ask workers about simple and objective performance outcomes (i.e., the absolute and relative performance outcomes) *and* when we ask workers about subjective outcomes that could reflect—as is often the case in self-evaluations and communications about one’s performance and ability in practice—workers’ beliefs about their absolute and relative performance as well as their subjective assessments of what constitutes a “poor performance.”

III. Evaluator Results

While the gender gap in confidence persists across the various self-evaluation questions we asked workers (as just shown in Section II), recall that our *Evaluator Study* focuses only on our main self-evaluation question (and we return to the other self-evaluation questions later in Section VA).

A. Results from the Baseline Treatment of the Evaluator Study

Table 2 presents our main results on evaluators’ beliefs, taken from the *Baseline* treatment of the *Evaluator Study*.

Column 1 (Prior) of Table 2 shows the evaluators’ prior beliefs—before they learn any information on workers’ self-evaluations—about the likelihood that workers have a poor performance. According to their priors, evaluators believe that there is a 42.97 percent chance of a female worker having a poor performance and there is a 39.08 percent chance of a male worker having a poor performance. That is, evaluators believe that female workers are 3.89 percentage points more likely to have a poor performance than male workers. While this expected performance gap is statistically significant (panel A), the expected performance gap is ultimately small and statistically indistinguishable from the true performance gap of 1.74 percentage points (panel B).

Column 2 (Overconfidence) of Table 2 shows evaluators’ beliefs about the likelihood that workers are overconfident. Evaluators believe men are much more likely to be overconfident: men are expected to be 8.25 percentage points significantly more likely than women to believe that they have a good performance when considering workers who actually have a poor performance (panel A). Nonetheless, this expected gender gap in overconfidence is significantly underestimated by 15.46 percentage points (panel B).

Column 3 (Underconfidence) of Table 2 shows the evaluators’ beliefs about the likelihood that workers are underconfident. Evaluators believe women are much more likely to be underconfident: women are expected to be 10.07 percentage points significantly more likely than men to believe they have a poor performance when considering workers who actually have a good performance (panel A). Nonetheless, this expected gender gap in underconfidence is significantly underestimated by 12.59 percentage points (panel B).

Column 4 (Implied Bayesian Posteriors) of Table 2 presents evaluators’ *implied Bayesian posterior beliefs*, which we define to equal what Bayesian evaluators would believe is the likelihood that a worker has a poor performance after they are provided with the information on workers’ self-evaluations. As detailed in online

TABLE 2—EVALUATORS' BELIEFS IN THE *BASILINE* TREATMENT OF THE *EVALUATOR STUDY*

	Prior (1)	Over- confidence (2)	Under- confidence (3)	Implied Bayesian Posterior (4)	Posterior (5)
<i>Panel A. Evaluators' beliefs</i>					
$B(F)$	42.97	39.86	55.68	43.83	61.85
$B(M)$	39.08	48.11	45.61	40.07	51.36
Δ	3.89	−8.25	10.07	3.77	10.49
SE of Δ	(1.87)	(2.27)	(2.06)	(1.87)	(1.78)
<i>Panel B. Evaluators' beliefs – truth</i>					
$B(F) - \text{Truth}(F)$	−6.56	24.51	−19.12	−5.70	12.32
$B(M) - \text{Truth}(M)$	−8.71	9.05	−6.53	−7.72	3.57
$\Delta - \text{Truth}(\Delta)$	2.15	15.46	−12.59	2.03	8.75
SE of $\Delta - \text{Truth}(\Delta)$	(1.87)	(2.27)	(2.06)	(1.87)	(1.78)
Observations	402	402	402	402	402
$\text{Truth}(F)$	49.53	15.35	74.80	49.53	49.53
$\text{Truth}(M)$	47.79	39.06	52.14	47.79	47.79
$\text{Truth}(\Delta)$	1.74	−23.70	22.65	1.74	1.74

Notes: Standard errors are robust and shown in parentheses. See online Appendix Table A.5 for definitions of evaluators' beliefs. For the evaluator belief noted in the column, panel A presents the average belief about female workers (see $B(F)$), the average belief about male workers (see $B(M)$), the difference in these averages (see Δ), and the standard error on the difference in these averages (see SE of Δ). For the evaluator belief noted in the column, panel B presents the average belief about female workers demeaned by the true value for female workers (see $B(F) - \text{Truth}(F)$), the average belief about male workers demeaned by the true value for male workers (see $B(M) - \text{Truth}(M)$), the difference in these demeaned averages (see $\Delta - \text{Truth}(\Delta)$), and the standard error on the difference in these demeaned averages (see SE of $\Delta - \text{Truth}(\Delta)$). At the bottom of the table, we provide corresponding true values for what evaluators' beliefs in panel A should be if evaluators are fully accurate when they are asked to provide beliefs about female workers (see $\text{Truth}(F)$) or male workers (see $\text{Truth}(M)$) as well as the difference in these values (see $\text{Truth}(\Delta)$). When considering evaluators' prior, implied Bayesian posterior, and posterior beliefs, we define these truth values as the actual likelihood of a randomly selected male/female worker having a poor performance. When considering evaluators' overconfidence and underconfidence beliefs, we define these truth values as the actual likelihood of a randomly selected male/female worker being overconfident conditional on having a poor performance and being underconfident conditional on having a good performance (see equations E4 and E5, respectively, in online Appendix E.E3). Data are from the 402 participants in the *Baseline* treatment of the *Evaluator Study*.

Appendix E, we can calculate each evaluator's implied Bayesian posterior belief given the three evaluator beliefs discussed so far: an evaluator's prior belief, overconfidence belief, and underconfidence belief. This is because evaluators' overconfidence and underconfidence beliefs determine their beliefs about the accuracy of the workers' self-evaluation information and hence how much they should update their prior beliefs after learning it. The implied Bayesian posterior beliefs reveal that—according to Bayesian updating—evaluators should expect that female workers are 3.77 percentage points more likely to have a poor performance than male workers after learning the workers' self-evaluation information—an expected performance gap that is statistically significant (panel A) but small and statistically indistinguishable from the true performance gap of 1.74 percentage points (panel B). This results from the fact that, in our data and as detailed in online Appendix Section E.E4, evaluators believe that workers are sufficiently miscalibrated in their self-evaluations such that a Bayesian evaluator would update very little from this information.

Another way to summarize the implied Bayesian posterior beliefs is as follows: according to evaluators' implied Bayesian posterior beliefs, the confidence gap should not be contagious. That is, if evaluators are Bayesian, the expected performance gap—after being provided with information on workers' self-evaluations—should be small and statistically indistinguishable from the true performance gap, even though this self-evaluation information conveys a large confidence gap. However, an examination of evaluators' posterior beliefs shows that this is not the case.

Specifically, column 5 (Posteriors) of Table 2 presents evaluators' posterior beliefs and shows that—unlike their prior beliefs and unlike their implied Bayesian posterior beliefs—evaluators' posterior beliefs do not reflect a small-to-nonexistent expected performance gap. Rather, after learning about more optimistic self-evaluations from male workers or more pessimistic self-evaluations from female workers, evaluators expect a substantial and statistically significant performance gap. They expect that female workers are 10.49 percentage points more likely to have a poor performance than male workers. This expected performance gap is both statistically significant (panel A) and substantially larger than the true performance gap of 1.74 percentage points (panel B). Indeed, this expected performance gap is more than 8.75 percentage points significantly larger than—or more than six times larger than—the true performance gap. In addition, when comparing priors to posteriors, the expected performance gap significantly increases by 6.61 percentage points.¹⁸

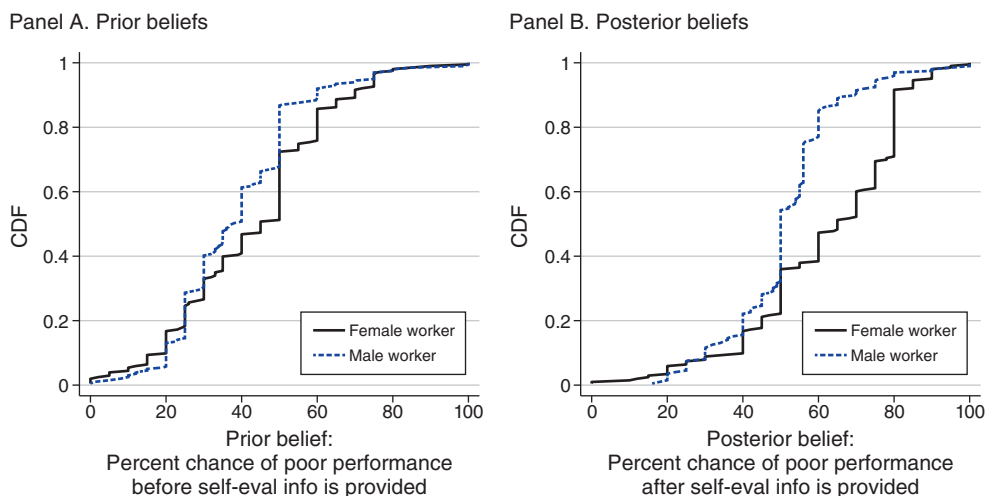
In summary, the confidence gap—conveyed via the gender gap in self-evaluations—exacerbates the expected performance gap, even though it should not if evaluators were Bayesians and even though evaluators expect a confidence gap (more on this in Section IVC). This contagious confidence gap results in overly pessimistic beliefs about women relative to men, as also evident by the distributions of prior beliefs and posterior beliefs shown in Figure 1 (see also online Appendix Figure B.1 for histograms).

B. Results from Attention and Calculation Treatments

One hypothesis as to why evaluators fail to accurately account for the confidence gap in the *Baseline* treatment—detailed above in Section IIIA—relates to an “attention” problem. For instance, since evaluators' overconfidence and underconfidence beliefs do reveal an expected gender gap in confidence, it could be that evaluators are simply *inattentive* to—but not unaware of—the influence of gender in self-evaluations when providing their posterior beliefs. This hypothesis could be enabled by the fact that, in the *Baseline* treatment, we elicit evaluators' overconfidence and underconfidence beliefs only *after* they provide their posterior beliefs. Thus, to investigate the attention problem via a light-touch intervention, we ran the *Attention* treatment. The *Attention* treatment elicits evaluators' overconfidence and underconfidence beliefs *before* they provide their posterior beliefs.¹⁹ Compare

¹⁸This 6.61 percentage point increase is statistically significant ($p < 0.01$) when regressing *prior-posterior* on an indicator for beliefs about female workers, with robust standard errors.

¹⁹Specifically, in the *Baseline* treatment, we elicit evaluators' beliefs in the following order: (i) their prior beliefs, (ii) their posterior beliefs, and then (iii) their overconfidence and underconfidence beliefs. By contrast, in

FIGURE 1. EVALUATORS' BELIEFS IN *BASILINE* TREATMENT

Note: Graphs show CDFs of the noted evaluators' beliefs from the *Baseline* treatment of the *Evaluator Study*.

online Appendix Figures A.1 and A.2 for a visual representation of this change between the *Baseline* treatment and *Attention* treatment.

Table 3 directly compares the *Baseline* treatment to the *Attention* treatment and shows that—for all evaluator beliefs—the expected performance gap is not significantly different between the *Baseline* treatment and *Attention* treatment.²⁰ The coefficient estimates on Δ reproduce the expected performance gap in the *Baseline* treatment, while the coefficient estimates on $\Delta \times \textit{Attention}$ show how the expected performance gap changes in the *Attention* treatment relative to the *Baseline* treatment. The coefficient estimates on $\Delta \times \textit{Attention}$ are small and never statistically significant. Thus, the *Attention* treatment does not significantly reduce the extent to which evaluators' posterior beliefs indicate an expected performance gap.

Another hypothesis for evaluators' failure to accurately account for the confidence gap relates to a “calculation” problem. For example, evaluators may be unable or unwilling to do the necessary calculations and Bayesian updating required to accurately account for the confidence gap. Thus, to investigate the effectiveness of a more extreme intervention that may help evaluators overcome any difficulty with Bayesian updating, we turn to the *Calculation* treatment. Like the *Attention* treatment, the *Calculation* treatment elicits evaluators' overconfidence and underconfidence beliefs *before* eliciting their posterior beliefs. In addition, the *Calculation* treatment uses evaluators' overconfidence and underconfidence beliefs—along with their prior beliefs—to inform evaluators of their implied Bayesian posteriors before

the *Attention* treatment, the order changes to be the following: (i) their prior beliefs, (ii) their overconfidence and underconfidence beliefs, and then (iii) their posterior beliefs.

²⁰ Following the structure of Table 2, online Appendix Table B.2 presents the results for the *Attention* treatment. Evaluators' beliefs in the *Attention* treatment are very similar to those in the *Baseline* treatment.

TABLE 3—EVALUATORS' BELIEFS IN THE *BASELINE*, *ATTENTION*, AND *CALCULATION* TREATMENT OF THE *EVALUATOR STUDY*

	Prior (1)	Over- confidence (2)	Under- confidence (3)	Implied Bayesian Posterior (4)	Posterior (5)
<i>Panel A. Evaluators' beliefs</i>					
Δ	3.89 (1.87)	−8.25 (2.27)	10.07 (2.06)	3.77 (1.87)	10.49 (1.78)
$\Delta \times \textit{Attention}$	−0.47 (2.62)	3.65 (3.16)	−0.23 (2.93)	−0.23 (2.60)	0.36 (2.48)
$\Delta \times \textit{Calculation}$	−0.81 (2.61)	−1.17 (3.21)	1.66 (2.86)	−0.66 (2.56)	−5.57 (2.54)
<i>Panel B. Evaluators' beliefs – truth</i>					
Δ	2.15 (1.87)	15.46 (2.27)	−12.59 (2.06)	2.03 (1.87)	8.75 (1.78)
$\Delta \times \textit{Attention}$	−0.47 (2.62)	3.65 (3.16)	−0.23 (2.93)	−0.23 (2.60)	0.36 (2.48)
$\Delta \times \textit{Calculation}$	−0.81 (2.61)	−1.17 (3.21)	1.66 (2.86)	−0.66 (2.56)	−5.57 (2.54)
Observations	1,210	1,210	1,210	1,209	1,210
Condition fixed effects	Yes	Yes	Yes	Yes	Yes
<i>Truth</i> (Δ)	1.74	−23.70	22.65	1.74	1.74

Notes: Standard errors are robust and shown in parentheses. See online Appendix Table A.5 for definitions of evaluators' beliefs. For the type of evaluator belief noted in the column, panel A presents an OLS of evaluators' belief on (i) suppressed indicators (i.e., Condition FEs) for the *Baseline* treatment, the *Attention* treatment, and the *Calculation* treatment, as well as (ii) an indicator for being asked about female workers (Δ), an indicator for being asked about female workers interacted with the indicator for the *Attention* treatment ($\Delta \times \textit{Attention}$), and an indicator for being asked about female workers interacted with the indicator for the *Calculation* treatment ($\Delta \times \textit{Calculation}$). For the type of evaluator belief noted in the column, panel B presents an OLS of evaluators' beliefs demeaned by the true values on the same set of indicators as in panel A. At the bottom of the table, we provide corresponding true values for the difference in evaluators' beliefs about female and male workers if evaluators are fully accurate when they are asked about female and male workers (see the estimates *Truth*(Δ)). Data are from the 1,210 participants in the *Baseline*, *Attention*, or *Calculation* treatment of the *Evaluator Study*. Sample size differs slightly in column 4 as some evaluators' beliefs imply a Bayesian posterior that is undefined.

they provide their posterior beliefs. See online Appendix Figure A.3 for a visual representation of this treatment.

Table 3 directly compares the *Baseline* treatment to the *Calculation* treatment and reveals one set of significant differences: according to evaluators' posteriors, the expected performance gap is significantly smaller in the *Calculation* treatment than in the *Baseline* treatment (column 5 of panel A) and significantly more accurate in the *Calculation* treatment than in the *Baseline* treatment (column 5 of panel B), while there are no significant differences in other beliefs.²¹ Thus, helping evaluators to update in a Bayesian manner in response to information on workers' self-evaluations significantly reduces the extent to which evaluators expect a performance gap.

²¹ Following the structure of Table 2, online Appendix Table B.3 presents the results for the *Calculation* treatment. Evaluators' beliefs in the *Calculation* treatment are similar to those in the *Baseline* and *Attention* treatment with one notable exception. While evaluators' posterior beliefs in the *Calculation* treatment indicate that they expect a performance gap, this expected performance gap is only *marginally* significantly different than the true gap and is noticeably smaller than what was observed in the other two treatments.

Given the effectiveness of the *Calculation* treatment, a natural question relates to the extent to which the *Calculation* treatment induces a sort of experimenter demand effect or social desirability bias. It could be the case that social pressure—whether from the experimenter, colleagues, or others—is a crucial component in encouraging individuals to accurately account for gender differences in confidence. It could also be the case that teaching individuals about Bayesian updating is somewhat inseparable from conveying to individuals how they *should* form their beliefs. Thus, while this type of experimenter demand effect or “teaching” could contribute to the results in the *Calculation* treatment, we leave open the possibility that this is a feature, not a bug. We also note that exploring other types of calculation interventions and assistance in forming Bayesian posteriors, including ones that would be more subtle, is an interesting avenue for future work.

Regardless, we provide three pieces of evidence that point against the relevance of experimenter demand effects or social desirability bias in our *Calculation* treatment. First, the majority of participants (61 percent) in the *Calculation* treatment report a posterior belief that differs from their implied Bayesian posterior belief, which shows that most participants are not simply reporting back the number that is suggested to them. Second, our results persist when only considering this 61 percent of participants.²² Third, as will become evident in Section IIIC, we will be able to show that—to the extent experimenter demand effects or social desirability bias drive the effectiveness of the *Calculation* treatment—this is not specific to gender (i.e., it is not specific to a potentially sensitive topic). Even in the *Unknown Gender* treatments, the *Calculation* treatment proves effective.

C. Results from Unknown Gender Treatments

To further investigate the calculation problem and understand the extent to which our results are specific to gender, we ran three additional treatments in which the gender of workers is not known. Specifically, for $X \in \{\text{Baseline}, \text{Attention}, \text{Calculation}\}$, the X , *Unknown Gender* treatment is the same as the X treatment except that instead of providing beliefs about male or female workers, evaluators provide beliefs about “group-1” or “group-2” workers. We tell evaluators that a worker is assigned to group-1 or group-2 based on how they answered a question in our follow-up survey, but we do not tell evaluators what this follow-up question is. In practice, we use the gender question from the follow-up survey, so group-1 workers are exactly the same set as our male workers and group-2 workers are exactly the same set as our female workers. This maintains the confidence gap between these two groups of workers while allowing differences between evaluators’ posterior beliefs and their implied Bayesian posterior beliefs to reflect failures in Bayesian updating but not any gender-specific biases.

Following the structure of Table 2, online Appendix Tables B.4–B.6 separately present the results from each of the three *Unknown Gender* treatments. There are three main takeaways. First, according to their prior beliefs and as one would expect

²²For the 61 percent of evaluators with differing posterior and implied Bayesian posterior beliefs in the *Calculation* treatment, evaluators’ posterior beliefs indicate an expected performance gap of 6.37 percentage points, which remains smaller than the expected gap of 10.49 percentage points in the *Baseline* treatment.

given the lack of information provided about group-1 and group-2 workers, evaluators in each treatment do not expect a performance gap. Second, evaluators in each treatment directionally, and sometimes to a statistically significant degree, expect that group-1 (male) workers are more likely to be overconfident conditional on a poor performance and that group-2 (female) workers are more likely to be underconfident conditional on a good performance. This demonstrates that—even without information on gender—evaluators quite reasonably believe that a group of workers is relatively more underconfident and relatively less overconfident when they learn that 80 percent of workers in that group believe they have a poor performance compared to when they learn that 56 percent of workers in that group believe they have a poor performance. Third, the confidence gap again results in overly pessimistic beliefs about women relative to men: according to their posterior beliefs, evaluators in each treatment expect that group-2 (female) workers are significantly more likely to have a poor performance than group-1 (male) workers.²³ Thus, the confidence gap is contagious even when it can only reflect a calculation problem about arbitrary groups and cannot reflect discriminatory motives or differences in priors. Indeed, the posterior beliefs in these *Unknown Gender* treatments are statistically indistinguishable from the posterior beliefs in the comparable treatments in which gender is known. Specifically, each column in Table 4 presents the posterior beliefs from a pair of treatments that compares the *X* treatment and the *X, Unknown Gender* treatment for $X \in \{\text{Baseline}, \text{Attention}, \text{Calculation}\}$. Across all three pairs of treatments, Table 4 reveals no significant differences in posteriors in the *Unknown Gender* treatments compared to those where gender is known (see the coefficients on $\Delta \times \text{Unknown Gender}$).²⁴

IV. Heterogeneity

To provide further insight into our results, we now turn to a series of additional results that are facilitated via heterogeneity analyses. For conciseness, we will focus on evaluators' posterior beliefs from the *Baseline* treatment, while showing how they are similar in the *Attention* treatment and indicate less of an expected performance gap between men and women in the *Calculation* treatment.

A. Are Our Results Driven by Evaluators Who Exhibit Other Cognitive Biases?

Motivated by the evidence in support of the calculation problem, one might expect a correlation between our results and well-known cognitive biases. To investigate this, we incentivize evaluators to correctly answer five additional questions

²³In all treatments, evaluators' posterior beliefs are significantly different than the truth. In addition, again pointing to the role of the calculation problem, evaluators' posterior beliefs are significantly different than their implied Bayesian posteriors in the *Baseline*, *Unknown Gender* and *Attention*, *Unknown Gender* treatment, but not in the *Calculation*, *Unknown Gender* treatment. Also, see online Appendix Table B.7 to compare the beliefs across the three *Unknown Gender* treatments.

²⁴If we consider evaluators' other beliefs, only two small differences arise. First, while evaluators' priors (and sometimes their posteriors) indicate that they expect a small performance gap when the worker gender is known, this is no longer the case when worker gender is unknown. Second, while evaluators' confidence beliefs indicate that they expect men to be significantly more overconfident and women to be more significantly more underconfident when worker gender is known, this is less true when worker gender is unknown.

TABLE 4—EVALUATORS' POSTERIOR BELIEFS ABOUT WORKERS ACCORDING TO WHETHER OR NOT THEY ARE IN AN *UNKNOWN GENDER* TREATMENT OF THE *EVALUATOR STUDY*

Evaluators' posterior beliefs in X and X , <i>Unknown Gender</i> condition given $X =$			
	<i>Baseline</i> (1)	<i>Attention</i> (2)	<i>Calculation</i> (3)
<i>Panel A. Evaluators' beliefs</i>			
Δ	10.49 (1.78)	10.85 (1.73)	4.92 (1.81)
$\Delta \times \text{Unknown gender}$	0.57 (2.40)	-0.29 (2.45)	-0.05 (2.53)
<i>Panel B. Evaluators' beliefs - truth</i>			
Δ	8.75 (1.78)	9.11 (1.73)	3.18 (1.81)
$\Delta \times \text{Unknown gender}$	0.57 (2.40)	-0.29 (2.45)	-0.05 (2.53)
Observations	807	795	798
Condition fixed effects	Yes	Yes	Yes
<i>Truth</i> (Δ)	1.74	1.74	1.74

Notes: Standard errors are robust and shown in parentheses. The data are from the X and X , *Unknown Gender* treatments noted in the columns. Panel A presents an OLS of evaluators' posterior beliefs on (i) suppressed indicators (i.e., Condition fixed effects) for the X treatment and the corresponding X , *Unknown Gender* treatment as well as (ii) an indicator for being asked about female workers (Δ) and an indicator for being asked about female workers interacted with the indicator for the X , *Unknown Gender* treatment ($\Delta \times \text{Unknown Gender}$). Panel B presents an OLS of evaluators' posterior beliefs demeaned by the true values on the same set of indicators as in panel A. At the bottom of the table, we provide corresponding true values for the difference in evaluators' beliefs about female and male workers if evaluators are fully accurate when they are asked about female and male workers (see the estimates *Truth*(Δ)). Data are from the 2,400 participants in the *Evaluator Study*, split across the three columns according to the relevant treatments.

at the end of the study: a standard Bayesian updating question, a question designed to detect base rate neglect (Kahneman and Tversky 1972a), and the three-question cognitive reflection test (CRT) (Frederick 2005), all presented in random order.²⁵

Table 5 presents results on how these measures correlate with the extent to which evaluators expect a performance gap, according to their posterior beliefs in the *Baseline* treatment in panel A, the *Attention* treatment in panel B, and the *Calculation* treatment in panel C. Counter to cognitive errors or general updating failures explaining our results, the expected performance gap is directionally larger for evaluators with higher cognitive ability scores (see the coefficients on $\Delta \times X$ in column 1) and is directionally smaller for evaluators who give a response farther from the Bayesian posterior in the Bayesian updating question (see the coefficients on $\Delta \times X$ in column 4). But, consistent with base rate neglect contributing to our results, the expected performance gap is directionally larger—sometimes significantly so—for evaluators who exhibit pure base rate neglect (see the coefficients on $\Delta \times X$ in column 2) or who give a response farther from the Bayesian posterior in the base rate neglect question (see the coefficients on $\Delta \times X$ in column 3). In addition, consistent with the *Calculation*

²⁵For full question text, see supplemental online Appendix Figures G.1.8–G.1.12.

TABLE 5—BY COGNITIVE ABILITY MEASURES: EVALUATORS' POSTERIOR BELIEFS ABOUT WORKERS IN *EVALUATOR STUDY* IN THE *BASILINE*, *ATTENTION*, AND *CALCULATION* TREATMENTS

$X =$	Evaluators' posterior beliefs			
	Demeaned CRT score (1)	Indicator for Base Rate Pure Neglect (2)	Demeaned error in base rate questions (3)	Demeaned error in Bayesian updating question (4)
<i>Panel A. Baseline treatment</i>				
Δ	8.64 (1.79)	7.77 (2.15)	8.76 (1.78)	8.91 (1.78)
$\Delta \times X$	0.96 (1.48)	3.15 (3.83)	0.35 (0.17)	-0.09 (0.09)
Observations	402	402	402	402
<i>Panel B. Attention treatment</i>				
Δ	9.02 (1.73)	6.55 (2.02)	9.14 (1.72)	9.11 (1.73)
$\Delta \times X$	0.90 (1.47)	7.91 (3.86)	0.36 (0.14)	-0.16 (0.09)
Observations	403	403	403	403
<i>Panel C. Calculation treatment</i>				
Δ	3.13 (1.79)	2.14 (2.08)	3.18 (1.81)	3.20 (1.80)
$\Delta \times X$	1.60 (1.51)	3.83 (4.18)	-0.04 (0.14)	-0.10 (0.08)
Observations	405	405	405	405
Suppressed X	Yes	Yes	Yes	Yes
$Truth(\Delta)$	1.74	1.74	1.74	1.74

Notes: Standard errors are robust and shown in parentheses. The data are from the *Baseline*, *Attention*, and *Calculation* treatments in panels A, B, and C, respectively. Each column presents an OLS of evaluators' posterior beliefs on (i) an indicator for being asked about female workers (Δ), (ii) a (suppressed) measure of X , and (iii) an interaction of the indicator in (i) and the measure of X . X is noted in each column and is: an evaluator's demeaned CRT score (out of three questions) in column 1, an indicator for whether the evaluator exhibited pure base rate neglect (where pure base rate neglect is consistent with ignoring the prior likelihood entirely) in column 2, the demeaned distance between the evaluator's answer and the Bayesian posterior in the base rate neglect question in column 3, and the demeaned distance between the evaluator's answer and the Bayesian posterior in the Bayesian updating bonus question in column 4. At the bottom of the table, we provide corresponding true values for the difference in evaluators' beliefs about female and male workers if evaluators are fully accurate when they are asked about female and male workers (see the estimates $Truth(\Delta)$).

treatment helping to eliminate the role of cognitive biases, we find that these relationships weaken in the *Calculation* treatment.

B. Are Our Results Driven by Evaluators with Certain Demographic Characteristics?

There are many reasons to expect our result to potentially correlate with different demographic groups. For instance, one may expect that, relative to female evaluators, male evaluators form more pessimistic posterior beliefs about women because they may have less experience with the confidence gap themselves or because of an in-group bias or discriminatory motives. This proves not to be the case. Table 6, which reproduces column 5 of Table 3 for male evaluators in column 1 and female evaluators in column 2, shows that male evaluators, if anything, hold *less*

TABLE 6—BY DEMOGRAPHICS: EVALUATORS' POSTERIOR BELIEFS ABOUT WORKERS IN EVALUATOR STUDY WHEN GENDER IS KNOWN

	Evaluators' posterior beliefs	
	Men (1)	Women (2)
Δ	9.31 (3.02)	11.56 (2.27)
$\Delta \times \text{Attention}$	-2.05 (4.18)	1.76 (3.09)
$\Delta \times \text{Calculation}$	-4.31 (4.08)	-5.83 (3.36)
Observations	507	669
Condition fixed effects	Yes	Yes
$\text{Truth}(\Delta)$	1.74	1.74

Notes: Standard errors are robust and shown in parentheses. The data are from the *Baseline*, *Attention*, and *Calculation* treatments for the group of evaluators noted in the column, specifically evaluators who: are men in column 1 and are women in column 2. Each column presents an OLS of evaluators' posterior beliefs on (i) suppressed indicators (i.e., Condition fixed effects) for the *Baseline*, *Attention*, and *Calculation* treatments as well as (ii) an indicator for being asked about female workers (Δ) and an indicator for being asked about female workers interacted with the indicator for the X treatment ($\Delta \times X$). At the bottom of the table, we provide corresponding true values for the difference in evaluators' beliefs about female and male workers if evaluators are fully accurate when they are asked about female and male workers (see the estimates $\text{Truth}(\Delta)$).

pessimistic posterior beliefs about women than female evaluators do. This adds to the evidence on situations in which believed gender differences are not only driven by men (see also Abrevaya and Hamermesh 2012; Babcock et al. 2017; Card et al. 2020; and Exley et al. 2022 for related results), and suggests the fact that evaluators' posteriors disfavor women in our study is not due to gender-specific bias or discrimination by male evaluators.²⁶

In addition, online Appendix Table C.4, which reproduces column 5 of Table 3 for various other demographics groups, shows that—regardless of whether we consider evaluators who are split according to their educational attainment, income, age or political affiliation—it is always the case that evaluators hold pessimistic posterior beliefs about women.

C. Do Our Results Persist for Evaluators Who Expect the Confidence Gap?

One could worry that our confidence elicitation is complicated or noisy or otherwise does not capture evaluators' true expectations about gender differences in confidence.²⁷ To provide additional evidence of the confidence gap being expected—and

²⁶There is, of course, a vast literature that often shows such an in-group bias does exist (see Tajfel et al. 1979; Chen and Li 2009; Chen and Chen 2011; Ioannou, Qi, and Rustichini 2016; Carlsson and Eriksson 2019 among many others). That said, even when an in-group bias is observed, future work may examine whether this in-group bias is specific to gender per se. Indeed, evidence from Coffman, Exley, and Niederle (2021) reveals that in-group preferences that are specific to gender and in-group preferences that are instead based on arbitrary groups can give rise to a similar pattern of discrimination in hiring decisions.

²⁷While we did not directly elicit confidence in one's beliefs, we find that over/underconfidence beliefs typically do not fall at 50 percent (see the histograms in online Appendix Figure B.2), which might have been an indicator of evaluators being entirely unsure about the confidence of men and women.

our results persisting among evaluators who expect the confidence gap—we can turn to data from two follow-up survey questions and to data from one of our additional study versions.

The two follow-up survey questions directly ask evaluators to categorize the relative confidence of men versus women. The first question asks evaluators to categorize the relative confidence of men versus women in general. While 46 percent of evaluators expect no gender difference in confidence, nearly all of remaining evaluators expect the confidence gap: 51 percent believe that women are less confident and only 3 percent believe that men are less confident. The second question asks specifically about confidence in math and science tasks, and similar results follow: while 42 percent of evaluators expect no gender difference in confidence, 51 percent believe that women are less confident while only 7 percent believe that men are less confident.

Online Appendix Table C.1 reproduces column 5 of Table 3 for each of these groups of evaluators. These results reveal that even evaluators who think women are less confident than men (columns 1 and 4) fail to account for the confidence gap: their posterior beliefs reveal a substantial and statistically significant expected performance gap. Similar results hold among evaluators who think there is no gender difference in confidence (columns 2 and 5). The results are noisier when restricting to the group of evaluators who think women are more confident than men (columns 3 and 6), likely due to the small sample size of this group.

In summary, most evaluators think that women are less confident than men—and almost no evaluators think the reverse is true—and our results persist even when we only consider evaluators who directly say that there is a confidence gap. In addition, as shown in online Appendix Tables D.10 and D.11, discussed in online Appendix D.D8, we can show—in a different study version in which we incentivize evaluators' confidence beliefs about both men and women—that our results persist among evaluators with incentivized overconfidence beliefs that directly indicate that they believe men are more overconfident than women, and among evaluators with incentivized underconfidence beliefs that directly indicate that they believe women are more underconfident than men.

D. Do Our Results Persist for Evaluators Who Think They Accurately Accounted for the Confidence Gap?

One might suspect that evaluators—if prompted to reflect on it—are aware that they did or did not accurately account for the gender gap in confidence in our study. To investigate this, we can turn to data from the following question that we ask in the follow-up survey of the known gender treatments: “When providing your predictions in this study, to what extent were you accounting for any gender differences in confidence?” 63 percent of evaluators answer “neither too little nor too much,” 14 percent of evaluators answer “slightly or far too much,” and 23 percent of evaluators answer “far or slightly too little.”

Online Appendix Table C.2 reproduces column 5 of Table 3 for each group of evaluators. Each group of evaluators expects a performance gap, according to their posterior beliefs. In addition, the expected performance gap is the *smallest* among evaluators who believe that they adjusted *too little* for gender differences

in confidence. Finally, when we instead ask evaluators whether they think employers—rather than themselves—accurately account for the confidence gap, similar results follow (see online Appendix Table C.3).

E. *Are Our Results Driven by Evaluators with Certain Other Beliefs?*

One may wonder whether our results are driven by evaluators who hold a particular set of initial beliefs. For instance, perhaps evaluators who seem most unsure about the chance that a male or female worker has a poor performance—and hence report a prior belief of 50 percent—are more susceptible to being influenced by information on workers' self-evaluations. This proves not to be the case (and we further note that only around 20 percent of evaluators have prior beliefs that fall right at 50 percent, as shown in online Appendix Figure B.1). For evaluators in the *Baseline* treatment, online Appendix Figure C.1 plots posterior beliefs as a function of evaluators' prior beliefs (panel A), overconfidence beliefs (panel B), underconfidence beliefs (panel C), and implied Bayesian posterior beliefs (panel D). These results make clear that evaluators' posterior beliefs disfavor women relative to men across the entire range of evaluators' other beliefs.²⁸

V. Robustness

In Section V, to investigate the robustness of our results, we turn to additional study versions.

A. *Are Our Results Robust to Evaluator Beliefs When Asked about Other Types of Performance Outcomes?*

As explained in Section IA, we chose to focus on *one* type of self-evaluation that we called our *main self-evaluation* question. To show that our results are robust to other types of self-evaluation questions—including those that relate to simpler performance outcomes—we ran two additional studies. Specifically, we recruited 400 new evaluators for the *Evaluator (Alternative Questions) Study* (see online Appendix Section D.D1 for results) and 400 new evaluators for the *Evaluator (Attention, Top Half) Study*.

In the *Evaluator (Alternative Questions) Study*, in addition to providing beliefs about the likelihood of a worker having poor performance in the manner defined in our main self-evaluation question (see online Appendix Table A.5), evaluators are also asked to provide beliefs about five other self-evaluations questions (see online Appendix Table A.6). This study is otherwise similar to the *Baseline* treatment of the *Evaluator Study*. As shown in online Appendix Table D.1, we find that—directionally, and almost always at a statistically significant level—our results hold across all of these performance outcomes: evaluators' priors indicate little to no gender differences, evaluators expect that male workers are more likely to be overconfident

²⁸ Online Appendix Figure C.2 shows that similar results follow in the *Attention* treatment. Online Appendix Figure C.3 shows that evaluators' prior beliefs and implied Bayesian posterior beliefs are more predictive in the *Calculation* treatment, which is perhaps related to the smaller expected performance gap in that treatment.

and female workers are more likely to be underconfident, their implied Bayesian beliefs imply no expected performance gap, but their posteriors indicate large and significant expected performance gaps. Specifically, evaluators' posteriors disfavor women according to two subjective classifications, and indicate that they expect that women are less likely to get 3+ questions right, less likely to get 5+ questions right, and less likely to perform in the top half. Only when asked about the percent chance of participants getting 7+ question right is a gender difference *not* expected—and this lack of a gender difference could reflect very few workers expecting to get 7+ questions right regardless of their gender (i.e., only 10 percent of male workers and 4 percent of female workers expected to get 7+ questions right).

While the *Evaluator (Alternative Questions) Study* shows the robustness of our results to other self-evaluations questions—including a simpler measure about whether a worker's performance is in the top half among other workers—the fact that we ask evaluators to provide beliefs about six self-evaluation questions could make the *Evaluator (Alternative Questions) Study* more complex than our main *Evaluator Study* in other ways. Thus, as an additional and important robustness check, we ran the *Evaluator (Attention, Top Half) Study* that *only* asks evaluators to provide beliefs about whether a worker's performance is in the top half among other workers and hence does *not* introduce any complexity by also asking evaluators to provide beliefs about other self-evaluation questions. In addition, motivated by a desire to mitigate the “attention problem” when considering evaluators' posterior beliefs, the *Evaluator (Attention, Top Half) Study* builds off of the *Attention*—rather than the *Baseline* treatment—of the *Evaluator Study*.

As shown in Table 7, even when evaluators are asked about a simple performance metric (i.e., being in the top half) and are in the *Attention* treatment, our results persist: evaluators' priors indicate no gender difference, evaluators expect that male workers are more likely to be overconfident and female workers are more likely to be underconfident, their implied Bayesian beliefs imply no expected performance gap, but their posteriors indicate a large and significant expected performance gap.

B. Are Our Main Results Robust to Asking Evaluators about the Full Distribution of Workers?

As explained in Section IC, we chose to focus on asking evaluators about workers with performances in the middle, or in the twenty-fifth to seventy-fifth percentile. To examine whether our results are robust to instead asking evaluators about the full pool of workers, we recruited 400 new evaluators for the *Evaluator (Full Distribution) Study* (see online Appendix Section D.D2 for results). This study asks evaluators about either a female worker who is randomly selected from the entire pool of female workers who completed the *Worker Study* or a male worker who is randomly selected from the entire pool of male workers who completed the *Worker Study*, but is otherwise identical to the *Baseline* treatment of the *Evaluator Study*. As shown in online Appendix Table D.2, we find that our results persist with similar magnitudes and with statistical significance: when providing beliefs about the full pool of workers, evaluators' priors indicate no gender difference, evaluators expect that male workers are more likely to be overconfident and female workers are more likely to be underconfident, their implied Bayesian beliefs imply no expected

TABLE 7—EVALUATORS’ BELIEFS IN THE *EVALUATOR (ATTENTION, TOP HALF) STUDY*

	Prior (1)	Over- confidence (2)	Under- confidence (3)	Implied Bayesian Posterior (4)	Posterior (5)
<i>Panel A. Evaluators’ beliefs</i>					
$B(F)$	47.15	38.05	46.42	45.36	40.52
$B(M)$	46.35	48.05	39.96	47.08	46.89
Δ	0.80	−10.00	6.46	−1.72	−6.38
SE of Δ	(1.82)	(2.19)	(2.04)	(1.85)	(1.73)
<i>Panel B. Evaluators’ beliefs – truth</i>					
$B(F) - \text{Truth}(F)$	0.81	24.41	−13.23	−0.98	−5.82
$B(M) - \text{Truth}(M)$	−2.06	15.74	0.62	−1.33	−1.52
$\Delta - \text{Truth}(\Delta)$	2.87	8.67	−13.85	0.35	−4.31
SE of $\Delta - \text{Truth}(\Delta)$	(1.82)	(2.19)	(2.04)	(1.85)	(1.73)
Observations	400	400	400	395	400
$\text{Truth}(F)$	46.34	13.64	59.65	46.34	46.34
$\text{Truth}(M)$	48.41	32.31	39.34	48.41	48.41
$\text{Truth}(\Delta)$	−2.07	−18.67	20.30	−2.07	−2.07

Notes: Standard errors are robust and shown in parentheses. Results follow the structure of Table 2. Data are from the 400 participants in the *Evaluator (Attention, Top Half) Study*. Note that the being in the “top half” meant that your score was greater than or *equal* to the scores of at least 50 percent of other participants, and these other participants are 50 randomly selected men and 50 randomly selected women.

performance gap, but their posteriors indicate a large and significant expected performance gap.

C. Are Our Results Robust to Evaluators with Hiring and Managerial Experience Providing Beliefs about Typical Job Candidates?

To investigate whether evaluators could better account for the confidence gap if they had more hiring and managerial experience and if they were asked about men and women who may be more “typical” of likely job candidates, we ran two additional studies. Specifically, we recruited 409 new evaluators for the *Baseline* treatment of the *Evaluator (Professional Evaluators) Study*, and 391 new evaluators for the *Baseline, Unknown Gender* treatment of the *Evaluator (Professional Evaluators) Study* (see online Appendix Section D.D4 for results). These studies are similar to the *Baseline* and *Baseline, Unknown Gender* treatments of the *Evaluator Study* aside from recruitment details. Specifically, the evaluators in these studies were recruited so that—according to self-reported data—they met the following two criteria: (i) they have experience in making hiring decisions (i.e., have been responsible for hiring job candidates) and (ii) they have experience in a management position.²⁹ In addition, rather than asking them about male and female workers recruited from Prolific, we asked them about people who are likely to be applying

²⁹Specifically, we use the internal screening questions on Prolific to recruit this sample. Participants’ answers to these questions are self-reported, and we cannot verify their work experience. That said, we note that the vast majority of Prolific participants do not meet these screening restrictions and that recent other papers who have used similar approaches include Huber and Huber (2020) and Saccardo and Serra-Garcia (2022). In our own follow-up survey, we can also confirm that 81 percent of these participants responded “yes” when asked a different but similar

for jobs in the near future: male and female workers who are undergraduate students at a large Midwestern university and expect to graduate in 2023. We thus also recruited 354 undergraduate students through Ohio State University for our *Worker (Undergraduates) Study* (see online Appendix Section D.D3 for results).

Following Table 1, online Appendix Table D.3 presents the results for these undergraduate students and confirms that the confidence gap persists for them.

Following Table 2, online Appendix Table D.4 presents the results for these “professional” evaluators in the *Baseline* treatment. Our main findings persist: these professional evaluators’ priors indicate no gender difference, they expect that male workers are more likely to be overconfident and female workers are more likely to be underconfident, their implied Bayesian beliefs imply no expected performance gap, but their posteriors indicate a large and significant expected performance gap.

Online Appendix Table D.5 presents parallel results for these professional evaluators in the *Baseline*, *Unknown Gender* treatment. The results in the *Baseline*, *Unknown Gender* treatment persist to a similar degree, showing that—even for professional evaluators and even when it is a calculation problem that cannot reflect discriminatory motives—the confidence gap is contagious.³⁰

D. Are Our Results Robust to Evaluators Gaining More Experience with Worker Self-Evaluations?

To investigate whether evaluators could better account for the confidence gap if they had more experience with the exact type of self-evaluations in our study, we recruited 406 new evaluators for the *Baseline* treatment of the *Evaluator (Extended) Study* (see online Appendix Section D.D5 for results).

The *Baseline* treatment of the *Evaluator (Extended) Study* is similar to the *Baseline* treatment of the *Evaluator Study* except for the “experience” stage. Specifically, after providing their prior beliefs—but before providing their posterior beliefs—evaluators gain “experience” by providing 20 beliefs about specific workers after learning each worker’s self-evaluation. For each of these 20 specific workers, evaluators are informed of the specific worker’s reported percent chance of having a poor performance and then are asked to report a belief about the percent chance of that specific worker having a poor performance (see supplemental online Appendix Figure I.5.6 for an example).³¹ As shown in online Appendix Table D.6, gaining experience with self-evaluations does not help evaluators to better account for the gender gap in confidence: experienced evaluators’ priors indicate no gender difference, they expect that male workers are more likely to be overconfident and female workers are more likely to be underconfident, their implied Bayesian beliefs imply a small expected performance gap that does not differ from the truth, but their posteriors indicate a large and significant expected performance gap.

question to Prolific screeners—i.e., when asked “Do you have any experience with decisions that relate to the hiring, pay, or promotion of employees or fellow colleagues?”

³⁰The expected gender gap in performance according to evaluators’ posterior beliefs—i.e., the estimates on Δ in columns 5 of online Appendix Tables D.4 and D.5—are statistically indistinguishable ($p > 0.1$).

³¹While we provide evaluators with information on how these specific workers answer the continuous Self-Evaluation Question 8C, the aggregate information we provide about the workers’ self-evaluations when eliciting our main posterior belief relates to how workers answered the binary Self-Evaluation Question 8B, consistent with our main *Evaluator Study*.

E. Are Our Results Robust to Beliefs about Specific Workers?

To investigate whether our results are robust to evaluator beliefs that pertain to a specific worker—after learning only that worker’s self-evaluation—we turn to the 20 worker-specific beliefs that evaluators provide in the experience stage of the *Baseline* treatment of the *Evaluator (Extended) Study*, described above in Section VD (again, see online Appendix Section D.D5 for results).

As shown in the northeastern region of online Appendix Figure D.1, there is some evidence that evaluators account for the confidence gap among the most pessimistic self-evaluations. For instance, when a worker reports an 80 percent chance of having a poor performance in their self-evaluation, the average evaluator believes there is a 74 percent chance of that worker having a poor performance if the worker is a man but only a 70 percent chance of that worker having a poor performance if that worker is a woman. Nonetheless, online Appendix Table D.7 shows that—even when asked about specific workers—evaluators’ posteriors indicate a large and significant expected performance gap.

F. Do Our Results Persist When Workers Face Strategic Incentives?

To investigate whether our results are robust to workers having strategic incentives that may encourage them to inflate their self-evaluations to potential evaluators—in a manner that is akin those in Exley and Kessler (2022) and as is the case in many settings outside of the laboratory—we ran two additional studies. Specifically, we recruited 387 new workers for the *Strategic Incentives* treatment of the *Worker Study* (see online Appendix Section D.D6 for results) and 394 new evaluators for the *Strategic Incentives* treatment of the *Evaluator (Extended) Study* (see online Appendix Section D.D7 for results).

In the *Strategic Incentives* treatment of the *Worker Study*, workers face strategic incentives because they earn more money if they are hired by an “employer” who learns their self evaluation. This study is otherwise similar to the *Worker Study*. Following Table 1, online Appendix Table D.8 presents the results for these workers and confirms that the confidence gap persists for them.

In the *Strategic Incentives* treatment of the *Evaluator (Extended) Study*, evaluators are provided with the self evaluations of these workers, and are informed of the workers’ strategic incentives. This study is otherwise similar to the *Baseline* treatment of the *Evaluator (Extended) Study*. Following Table 2, online Appendix Table D.9 presents the results for these evaluators and shows that our results persist: even when workers face strategic incentives, evaluators’ priors indicate no gender difference, they expect that male workers are more likely to be overconfident and female workers are more likely to be underconfident, their implied Bayesian beliefs imply no performance gap, but their posteriors indicate a large and significant expected performance gap.

G. Are Our Results Robust to Being Asked about Both Men and Women?

To investigate whether evaluators could better account for the confidence gap if they were making explicit comparisons between male and female workers—inspired

by prior work that suggests judgments are less reasoned when comparison information is lacking (Bohnet, van Geen, and Bazerman 2016)—we ran two additional studies. Specifically, we recruited 205 new evaluators for the *Joint Evaluations* treatment of the *Evaluator (Extended) Study* (see online Appendix Section D.D8 for results) and 195 new evaluators for the *Joint Evaluations, Strategic Incentives* treatment of the *Evaluator (Extended) Study* (see online Appendix Section D.D8 for results). In these treatments, we asked evaluators to provide beliefs about a male worker and a female worker *on the same decision screen*. These studies are otherwise similar to the *Baseline* treatment and *Strategic Incentives* treatment of the *Evaluator (Extended) Study*, respectively.

Following the same specifications as those in Table 2, online Appendix Table D.10 presents results for evaluators in the *Joint Evaluations* treatment and online Appendix Table D.11 presents results for evaluators in the *Joint Evaluations, Strategic Incentives* treatment. In both cases, our results persist: when providing joint evaluations—for workers who faced strategic incentives or otherwise—evaluators' priors indicate little to no gender differences, they expect that male workers are more likely to be overconfident and female workers are more likely to be underconfident, their implied Bayesian beliefs imply little to no expected performance gap, but their posteriors indicate a large and significant expected performance gap.

H. Are Our Results Robust to Considering Evaluators' Beliefs at the Individual Level?

Since the evaluators from our main *Evaluator Study* provide beliefs about only male or female workers as discussed in Section IC, those results do not allow us to classify evaluators—at the individual level—according to whether they expect female workers to be more, equally, or less likely to have a poor performance than male workers. But, our *Joint Evaluations* treatment and *Joint Evaluations, Strategic Incentives* treatment allow for such classifications.

Figure 2 presents the results from the *Joint Evaluations* treatment. When evaluators are classified according to their prior beliefs, shown via the light blue bars, we find that the percent of evaluators who think female workers are more, equally, or less likely to have a poor performance is 39 percent, 43 percent, and 18 percent respectively. But, the confidence gap causes a substantial increase—indeed a doubling—in the percent of evaluators who believe that female workers are more likely to have a poor performance than male workers. When evaluators are classified according to their posterior beliefs, shown via the black bars, the percent of evaluators who think female workers are more, equally, or less likely to have a poor performance is 79 percent, 11 percent, and 10 percent respectively. Thus, even when considering the individual-level results, the confidence gap is contagious. See online Appendix Figure D.2 for similar results from the *Joint Evaluations, Strategic Incentives* treatment.

I. Are Our Results Robust to Conveying Gender More Subtly?

To investigate whether our results are robust to conveying gender more subtly—and relatedly, to ensure that our results are not driven by experimenter demand

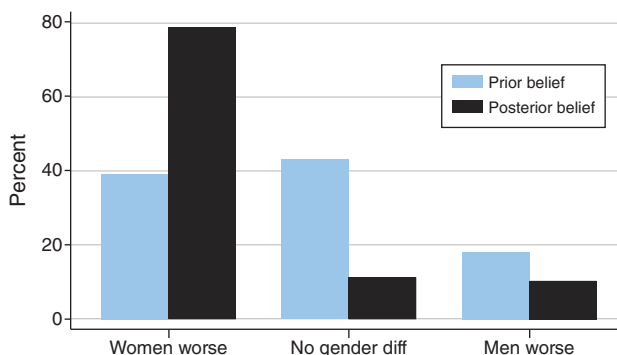


FIGURE 2. JOINT EVALUATIONS TREATMENT: CLASSIFYING EVALUATORS ACCORDING TO THEIR BELIEFS

Notes: This graph shows the percent of evaluators who, given their prior or posterior beliefs, believe that women—relative to men—are more, equally, or less likely to have a poor performance in the first, middle, and right pair of bars, respectively. Data are from the *Joint Evaluations* treatment of the *Evaluator (Extended) Study*.

effect or social desirability bias—we recruited 198 new evaluators for the *Evaluator (Additional Demographics) Study* (see online Appendix Section D.D9 for results). Specifically, in this study, we tell evaluators that their worker will be randomly drawn from a group of workers who work full time, are between 26 and 40 years old, live in the Southern region of the United States, have completed at least some college education, and are (wo)men.³² This study is otherwise similar to the *Baseline* treatment of the *Evaluator Study*.

Following Table 2, online Appendix Table D.12 presents results for the evaluators in the *Evaluator (Additional Demographics) Study* and shows that our results persist: evaluators' priors indicate no gender difference, they (directionally) expect that male workers are more likely to be overconfident and female workers are more likely to be underconfident, their implied Bayesian beliefs imply no expected performance gap, but their posteriors indicate a large and significant expected performance gap.

J. Are Our Results Robust to Situations Where Absolute Performance Information Is Known about the Workers?

To investigate the robustness of our results to a situation where absolute performance information is known, we recruited 198 new evaluators for the *Evaluator (Known Performance) Study* (see online Appendix Section D.D10 for results). Specifically, in this study, we tell evaluators that their worker will be randomly drawn from the group of male or female workers who got 5 questions right on the math and science test—ensuring their worker's *absolute performance* is known with certainty. This study is otherwise similar to the *Baseline* treatment of the *Evaluator Study*.

³²These demographics were modal in the *Worker Study*, with modal age being the modal generation. When comparing these groups of workers, male workers have a 43 percent chance of poor performance compared to 35 percent for female workers ($p = 0.51$); nevertheless, as with our prior result, these female workers report significantly more pessimistic self-evaluations: 77 percent of female workers in this group believe they have a poor performance while only 38 percent of male workers do.

Following Table 2, online Appendix Table D.13 presents the results from the *Evaluator (Known Performance) Study* and shows that our results persist: evaluators' priors indicate no gender difference, they (directionally) expect that male workers are more likely to be overconfident and female workers are more likely to be underconfident, their implied Bayesian beliefs imply no expected performance gap, but their posteriors indicate a large and significant expected performance gap.

We make one additional note on these results. One might wonder why evaluators in the *Evaluator (Known Performance) Study* update *at all* in response to the provided self-evaluation information about whether a worker believes they have a poor performance, given that evaluators know the worker got 5 questions right on the math and science test. If evaluators had a stable mapping from absolute performance outcomes (i.e., the number of questions the worker got right) to the subjective performance outcome of interest (i.e., whether the worker has a poor performance), then learning workers' self-evaluation information would not affect evaluators' beliefs. Alternatively, if evaluators' beliefs about subjective performance outcomes depend on more than the number of questions the worker got right—e.g., they also are influenced by others' views about what constitutes a poor performance—then such a stable mapping may not exist. That there need not be a stable mapping between absolute and subjective performance outcomes is one of the reasons we chose to focus on a subjective performance outcome in our main self-evaluation question. Ultimately, because of this, we can show that the confidence gap is contagious—causing overly pessimistic beliefs about women relative to men—even when women and men are known to have answered exactly the same number of questions right. More generally, since many factors could influence individuals' views of subjective performance outcomes (including absolute performance outcomes, other objective performance outcomes, other subjective criteria such as those relating to one's standards, or even confusion), these results suggest that the contagious confidence gap may arise even in situations in which ample information on a worker is known. We leave further investigation of this to future work.

VI. Conclusion

Through a series of experiments in which evaluators are incentivized to provide accurate beliefs, we document that evaluators *expect* a confidence gap, but they do not *account for* it. Specifically, we show that the confidence gap—conveyed via workers' self-evaluations about their performance on a math and science test—results in overly pessimistic beliefs about women relative to men. This “contagious” confidence gap arises even though it should not have if evaluators were Bayesian and even though the confidence gap is expected. Additional results support the interpretation of this contagious confidence gap reflecting more of a calculation problem, rather than an awareness or attention problem.

We see many important avenues for future work, four of which we mention here. One stream of future work may investigate ways to counter the contagious confidence gap, particularly since the confidence gap may be conveyed via self-evaluations in hiring, promotion, and pay decisions. Given the ineffectiveness of our *Attention* treatment and the fact that individuals expect the confidence gap, our results highlight how awareness or attention need not be sufficient. That said, future work may

reveal more effective attention interventions, which likely depend on the salience of the attention intervention and the context itself (e.g., an attention intervention in which employers reviewing job candidates view a pop-up window that says “remember that women are typically underconfident” may prove more effective than an attention intervention that elicits related beliefs as in our *Attention* treatment).³³ In addition, motivated by our results reflecting a calculation problem—notably including the effectiveness of the *Calculation* treatment and that similar results arise when evaluators are asked about arbitrary rather than gender-specific groups—our results suggest that a confidence gap, resulting from gender or other group differences, can cause non-Bayesian agents to be biased even absent any explicit discriminatory motives. This lends particular promise to strategies that help individuals overcome cognitive limitations, even when these limitations are not directly related to factors such as gender. For instance, while evaluators did not receive feedback in our experiment, future work may test the effectiveness of allowing evaluators to *learn* about their biases via iterative feedback. Also, given the positive correlation between the extent to which evaluators’ posteriors disfavor women and the extent to which they exhibit base rate neglect, future work may test the effectiveness of strategies that build off of insights from the broader literature on cognitive limitations and behavioral biases.³⁴ When strategies are uncovered that do work, important questions will also relate to the *relative* effectiveness of these strategies and the conditions under which they work.

A second stream of future work may explore the impact of removing gender information from applications and various types of evaluations (see Kolev, Fuentes-Medel, and Murray 2019). On one hand, in light of the literature on gender-specific backlash and discrimination more generally (Riach and Rich 2002; Rudman and Fairchild 2004; Bowles, Babcock, and Lai 2007; Rudman and Phelan 2008), the removal of gender information could prove helpful. On the other hand, the removal of gender information likely decreases the chance that employers can accurately account for gender differences in confidence, even if they are provided with the training and tools to do so.

A third stream of future work may explore how others form beliefs about men and women in settings in which the size, and magnitude, of the confidence gap—and the actual performance gap—vary due to any number of factors such as the relevance of stereotypes and the selection of individuals involved. Such future work may reveal: more situations in which the confidence gap is contagious and causes overly pessimistic beliefs about women relative to men, situations in which the confidence gap does not exist, and even situations in which the confidence gap causes overly *optimistic* beliefs about women relative to men. To better understand these situations, however, we hope future work considers how men and women’s beliefs about

³³ The effectiveness of attention interventions could be different in settings involving more free-form communication (Coffman, Flikkema, and Shurchkov 2019) or that require updating from the lack of information (Enke 2020; Charness, Oprea, and Yuksel 2022; Agan, Cowgill, and Gee 2023). Also, the effect of the intervention could depend on the gender of individuals selecting into the context, see Exley, Niederle, and Vesterlund (2020) for an example of how selection influences when it is a good idea to negotiate.

³⁴ As an example of this in the broader literature on non-Bayesian updating, Gonçalves, Libgober, and Willis (2021) show that individuals can fail to “unlearn” from signals that are retracted. It would be interesting to see how these results carry over to an environment with self-evaluations; that is, if evaluators update from self-evaluations, but then are *told* that the self-evaluations are biased, do evaluators sufficiently unlearn the self-evaluation?

themselves—often communicated through self-evaluations—affect the beliefs that others hold about men and women.

A fourth stream of future work may examine how these results extend beyond gender and to other biases. On extending beyond gender, since the confidence gap is contagious even when evaluators are asked about arbitrary rather than gender-specific groups, future work may naturally investigate whether similar results follow whenever individuals are asked about two groups as long as one of those groups has lower confidence. On extending to other biases, future work may investigate whether biases—driven by different sources than the confidence gap—also result in similar findings. For instance, future work may explore whether individuals expect that certain groups face discrimination but nonetheless fail to account for discrimination when evaluating those groups. This future work may also explore whether expecting a bias creates a false sense of confidence in one's ability to account for it, which may in turn hinder debiasing attempts. Indeed, as discussed in Section IVD, we find that posterior beliefs reveal expected gender gaps in performance that are, if anything, larger for individuals who think they accurately accounted or over-accounted for the confidence gap relative to those who think that they under-accounted for it.

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